

Chapter 8C

GI, GU, Intra-Abdominal, Sexually Transmitted, Hepatitis, and HIV-Linked Infection Patterns

Medvale Internal Medicine - Part VIII: Infectious Diseases

Why this matters clinically

A large fraction of inpatient and ambulatory infectious-disease decisions begin with the same question: where is the source? Gastrointestinal, hepatobiliary, genitourinary, pelvic, and sexually transmitted infections often share fever, leukocytosis, malaise, abdominal pain, nausea, dysuria, or nonspecific systemic illness. The diagnostic challenge is to localize the infection, decide whether the host can safely be treated as an outpatient, identify complications needing imaging or drainage, and choose empiric therapy that is broad enough for the syndrome but narrow enough to avoid resistance, toxicity, and *Clostridioides difficile* recurrence.

This chapter is a content-heavy original Medvale segment. The uploaded infection source provides coverage signals for viral hepatitis, botulism, intra-abdominal abscess, *C. difficile* colitis, lower UTI, pyelonephritis, prostatitis, genital warts, and chlamydia. Current guideline resources are used only to cross-check clinical safety anchors.

Core illness scripts

Syndrome	Fast illness script	Immediate decision point
Acute viral hepatitis	Fatigue, nausea, RUQ discomfort, jaundice, dark urine, marked AST/ALT elevation; exposures include food/water, blood, sex, injection use, birth country, and perinatal transmission.	Assess synthetic failure: INR, mental status, bilirubin, glucose, renal function. Fulminant hepatitis is liver failure until proven otherwise.
<i>C. difficile</i> colitis	Watery diarrhea after antibiotics, hospitalization, age, gastric acid suppression, IBD, or immunocompromise; may progress to ileus, toxic megacolon, shock.	Stop inciting antibiotic when possible; severity decides oral fidaxomicin/vancomycin versus fulminant regimen plus surgical awareness.
Uncomplicated cystitis	Dysuria, frequency, urgency, suprapubic pain, pyuria; usually no fever or flank pain.	Treat empirically in classic cases; culture if recurrent, complicated, pregnancy, male sex, failure, or atypical symptoms.
Pyelonephritis	Fever, chills, flank pain/CVA tenderness, nausea/vomiting, pyuria; may follow cystitis but may present without lower symptoms.	Decide outpatient oral therapy versus admission for sepsis, pregnancy, obstruction, inability to tolerate PO, renal transplant, or severe comorbidity.
Acute prostatitis	Fever, chills, pelvic/perineal pain, obstructive urinary symptoms, tender boggy prostate; often gram-negative rods.	Avoid vigorous prostate massage; if toxic, admit for IV therapy and evaluate for retention or abscess.
Chlamydial urethritis/cervicitis	Often asymptomatic; dysuria, urethral/cervical discharge, intermenstrual or postcoital bleeding; can ascend to PID.	Test with NAAT, treat patient and partners, and screen for gonorrhea/HIV/syphilis based on risk.

I. Viral hepatitis and hepatotropic infection logic

Hepatitis means inflammatory injury to hepatocytes. Infection is one cause, but the differential includes alcohol-associated hepatitis, drug-induced liver injury, ischemic hepatitis, autoimmune hepatitis, biliary obstruction, metabolic liver disease, and systemic illness. The first clinical sorting move is to decide whether the patient has acute hepatocellular injury, cholestatic injury, chronic hepatitis, or acute liver failure.

A hepatocellular pattern has AST and ALT out of proportion to alkaline phosphatase. A cholestatic pattern has alkaline phosphatase and direct bilirubin out of proportion to transaminases. Acute viral hepatitis can push AST/ALT above 1000 U/L, but so can ischemic hepatitis, acetaminophen toxicity, severe drug injury, and autoimmune hepatitis. Chronic hepatitis may have only mild transaminase elevation even while fibrosis progresses silently.

Virus	Transmission	Chronicity	Key diagnostic anchor	Clinical hooks
HAV	Fecal-oral: contaminated food/water, close contact, outbreaks	No chronic infection	IgM anti-HAV for acute infection; IgG anti-HAV for immunity	Abrupt febrile GI illness, jaundice; vaccine-preventable; supportive care

HBV	Blood, sex, perinatal exposure, injection equipment	Can become chronic; risk highest with perinatal infection	HBsAg = infection; anti-HBs = immunity; anti-HBc IgM = acute/window; HBeAg = high infectivity	Linked to PAN, membranous nephropathy, HCC; vaccine-preventable
HCV	Blood exposure, injection equipment; less often sexual/perinatal	Frequently chronic if untreated	HCV antibody screens exposure; HCV RNA confirms active infection	Often asymptomatic until fibrosis/cirrhosis; associated with mixed cryoglobulinemia
HDV	Requires HBV envelope; coinfection or superinfection	Can accelerate severe chronic liver disease	Anti-HDV or HDV RNA in HBV-infected patient	Think in severe HBV phenotype or high-risk exposure
HEV	Fecal-oral; waterborne; zoonotic in some settings	Usually no chronic infection except immunocompromised hosts	HEV IgM or RNA when suspected	Pregnancy can be severe; supportive in most immunocompetent patients

Acute hepatitis algorithm

1. Confirm pattern: AST/ALT predominant versus alkaline phosphatase/direct bilirubin predominant. Repeat if hemolyzed or inconsistent with presentation.
2. Screen severity: INR, bilirubin, albumin, glucose, creatinine, platelets, mental status. Encephalopathy plus INR elevation suggests acute liver failure.
3. Ask exposure timeline: food/water/travel, sexual exposure, injection drug use, needlestick, transfusion history, tattoos, incarceration, birth country, pregnancy, medications, supplements, alcohol, hypotension.
4. Order targeted serologies: HAV IgM; HBV panel; HCV antibody with reflex RNA; HEV testing when epidemiology fits; HIV testing when risk is present.
5. Treat syndrome and transmission risk: supportive care for HAV/HEV, public-health counseling, vaccination of contacts when appropriate, and antiviral referral for chronic HBV/HCV or severe disease.

Treatment principles

HAV and most HEV infections are managed supportively with hydration, antiemetics, avoidance of alcohol and hepatotoxins, and monitoring for liver failure. HBV is prevented by vaccination and post-exposure prophylaxis when indicated; chronic HBV treatment decisions depend on HBV DNA level, ALT, fibrosis/cirrhosis, HBeAg status, age, pregnancy, and immunosuppression plans. HCV is now usually curable with direct-acting antivirals; modern guidance strongly favors treating nearly all patients with acute or chronic active HCV unless life expectancy is too limited to benefit from cure.

Common traps

- Do not interpret isolated HCV antibody as active HCV infection; confirm with HCV RNA.
- Do not miss the HBV window period: HBsAg may be negative and anti-HBs not yet positive; anti-HBc IgM is the clue.
- Do not assume AST/ALT magnitude equals prognosis. INR, encephalopathy, renal failure, acidosis, and hypoglycemia are more urgent danger signals.
- Do not start immunosuppressive biologic therapy, high-dose steroids, or chemotherapy without considering HBV reactivation risk.

II. Foodborne neurotoxins and botulism

Botulism is not a typical gastroenteritis syndrome. It is a toxin-mediated neurologic emergency caused by botulinum neurotoxin, which prevents acetylcholine release at the neuromuscular junction. The classic pattern is acute, symmetric, descending flaccid weakness with cranial nerve involvement: blurred vision, diplopia, ptosis, dysarthria, dysphagia, dry mouth, and eventually respiratory failure. Sensation is preserved and fever is usually absent unless another process coexists.

Feature	Botulism	Guillain-Barre	Myasthenia gravis	Tick paralysis
Weakness pattern	Descending, symmetric, cranial nerves early	Ascending, often legs first	Fluctuating ocular/bulbar/proximal weakness	Ascending weakness after tick attachment
Pupils	Often dilated or poorly reactive	Usually normal	Normal	Usually normal
Sensation	Preserved	May have paresthesias	Preserved	Preserved
Reflexes	May be decreased	Decreased/absent	Usually preserved	Decreased

Management trigger	Antitoxin immediately; monitor vital capacity	IVIG/plasma exchange; respiratory monitoring	Acetylcholinesterase inhibitors/immunotherapy depending context	Remove tick; supportive
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If botulism is suspected, do not wait for confirmatory testing before involving public health and administering antitoxin. Management is airway-centered: serial forced vital capacity, negative inspiratory force when available, ICU monitoring for bulbar dysfunction, and early intubation when respiratory mechanics worsen.

III. Intra-abdominal infection, abscess, and source control

Intra-abdominal infection is less about choosing the perfect antibiotic and more about deciding whether infection is contained or uncontrolled. A contained abscess may require percutaneous drainage plus antibiotics. A perforated viscus, ischemic bowel, uncontrolled biliary obstruction, appendiceal rupture, or anastomotic leak requires source control. Antibiotics cannot sterilize a poorly drained abscess, dead tissue, infected hardware, or a blocked duct.

Syndrome	Usual organisms	Diagnostic clues	Management logic
Intra-abdominal abscess	Polymicrobial: Enterobacterales, enterococci, anaerobes	Fever, pain, leukocytosis after perforation, surgery, pancreatitis, diverticulitis, appendicitis, pelvic infection; CT localizes collection	Drain if feasible; broad gram-negative and anaerobic coverage; tailor to cultures
Spontaneous bacterial peritonitis	Enteric gram-negative rods, streptococci, enterococci	Cirrhosis with ascites plus fever, abdominal pain, encephalopathy, renal dysfunction; PMN $\geq$ 250 cells/mm ³	Diagnostic paracentesis before antibiotics when possible; ceftriaxone-like therapy; albumin in selected high-risk patients
Secondary peritonitis	Polymicrobial with anaerobes	Rigid abdomen, high fever, sepsis, very high ascitic protein/LDH or multiple organisms	Urgent imaging/surgery; source control mandatory
Biliary sepsis	Enteric gram-negatives, enterococci, anaerobes in severe cases	RUQ pain, fever, jaundice; cholangitis may include hypotension/altered mental status	Antibiotics plus biliary decompression when obstruction/cholangitis

Antibiotic coverage follows severity and acquisition context. Community-acquired mild-to-moderate intra-abdominal infection often requires Enterobacterales and anaerobic coverage. Healthcare-associated infection, severe sepsis, prior resistant organisms, transplant, or recent broad antibiotic exposure may require broader empiric therapy while cultures are pending. De-escalation after source control is part of treatment, not an afterthought.

IV. Clostridioides difficile infection

C. difficile colitis is toxin-mediated disruption of the colon after the normal microbiome is injured. Antibiotics are the classic trigger, but risk also rises with hospitalization, age, immunosuppression, inflammatory bowel disease, tube feeding, gastric acid suppression, and prior CDI. The syndrome ranges from mild watery diarrhea to fulminant colitis with ileus, toxic megacolon, shock, and perforation.

CDI category	Clinical anchors	Treatment framework
Initial nonfulminant episode	Diarrhea plus positive stool toxin/NAAT in a compatible patient; no shock, ileus, or megacolon	Fidaxomicin preferred when feasible; oral vancomycin remains acceptable. Avoid testing formed stool or asymptomatic patients.
Severe nonfulminant	WBC $\geq$ 15,000 cells/uL or creatinine $\geq$ 1.5 mg/dL in many frameworks	Fidaxomicin or oral vancomycin; monitor hydration, renal function, leukocytosis, abdominal exam.
Fulminant	Hypotension/shock, ileus, toxic megacolon, severe abdominal distention, lactate elevation	High-dose oral/NG vancomycin plus IV metronidazole; rectal vancomycin if ileus; early surgical consultation.
First recurrence	Symptoms return after successful initial treatment	Fidaxomicin, vancomycin taper/pulse, or other recurrence strategy depending prior regimen and access.
Multiple recurrences	Repeated relapses, high morbidity	Fidaxomicin/taper strategies, bezlotoxumab in selected high-risk patients, fecal microbiota-based therapy when appropriate.

Diagnostic testing should be restricted to patients with clinically significant diarrhea, usually at least 3 unformed stools in 24 hours without a better explanation. NAAT is sensitive but can detect colonization; toxin assays are more specific for active toxin production but less sensitive. Persistent PCR positivity after treatment does not prove active disease; do not test for cure.

Watery diarrhea after antibiotics: CDI reasoning

1. Confirm true diarrhea and timeline. Exclude laxatives, tube feeds, bowel prep, overflow diarrhea, IBS flare, and viral gastroenteritis when obvious.
2. Determine severity: vitals, WBC, creatinine, albumin, abdominal distention, ileus, lactate, peritonitis, megacolon.
3. Stop unnecessary inciting antibiotics and acid suppression. Use contact precautions and soap-and-water hand hygiene because spores persist.
4. Treat according to severity. If fulminant features appear, add surgical awareness early rather than after perforation or shock becomes irreversible.
5. For recurrence, treat relapse as a distinct clinical problem: choose recurrence-focused therapy and reassess ongoing risk factors.

V. Lower urinary tract infection

A UTI is not simply bacteriuria. The clinical diagnosis depends on symptoms plus supportive urinalysis. Many older adults, catheterized patients, and patients with chronic urinary abnormalities have bacteriuria without tissue invasion. Treating asymptomatic bacteriuria usually causes harm: resistance, adverse drug effects, and CDI. Important exceptions include pregnancy and selected urologic procedures with mucosal bleeding risk.

Category	Definition in practice	Common pathogens	Testing	Treatment logic
Uncomplicated cystitis	Nonpregnant adult woman with dysuria/frequency/urgency and no fever, flank pain, structural urinary disease, immunocompromise, or systemic illness	E. coli most common; also S. saprophyticus, Klebsiella, Proteus, Enterococcus	UA often enough in classic symptoms; culture if atypical, recurrent, treatment failure	Nitrofurantoin, TMP-SMX if local resistance acceptable, fosfomycin. Avoid fluoroquinolone when narrower options fit.
Complicated UTI	Male sex, pregnancy, obstruction, stone, catheter, renal transplant, diabetes with severe/systemic disease, immunocompromise, resistant organism risk, abnormal tract	Broader gram-negative rods, Enterococcus, Pseudomonas, Candida in selected settings	Urine culture before therapy when possible	Empiric therapy guided by severity, prior cultures, local resistance; longer course and source correction.
Asymptomatic bacteriuria	Positive culture without urinary symptoms or systemic signs attributable to urinary source	Varies	Do not screen most groups	Treat pregnancy and before selected invasive urologic procedures; otherwise usually do not treat.
Catheter-associated bacteriuria	Bacteriuria common with duration of catheterization	Polymicrobial and resistant organisms common	Culture from a freshly placed catheter when treating	Remove/replace catheter; treat only symptomatic infection or systemic illness attributable to urinary source.

Urinalysis interpretation

UA finding	Meaning	Trap
Leukocyte esterase	Suggests pyuria from WBCs	Pyuria is inflammation, not proof of bacterial infection; stones, STIs, catheter irritation, interstitial nephritis, and TB can cause pyuria.
Nitrite	Suggests nitrate-reducing gram-negative rods such as many Enterobacterales	Negative nitrite does not rule out UTI; Enterococcus, Staph saprophyticus, frequent voiding, and low bacterial burden can be nitrite-negative.
Squamous epithelial cells	Contamination from skin/vaginal epithelium	Culture results from contaminated specimens may lead to overtreatment.
WBC casts	Inflammation within kidney tubules/interstitium	Suggests pyelonephritis or interstitial nephritis rather than simple cystitis.
Hematuria	Can occur with cystitis or pyelonephritis	Persistent hematuria after infection clears needs evaluation for stone, malignancy, glomerular disease, or structural lesion.

The most useful clinical split is lower versus upper tract infection. Fever, chills, flank pain, CVA tenderness, nausea/vomiting, hypotension, or leukocytosis moves the diagnosis toward pyelonephritis or complicated infection. Pure dysuria with discharge or sexual exposure should raise urethritis/cervicitis rather than cystitis, especially when UA shows pyuria but culture is negative.

VI. Pyelonephritis and complicated upper UTI

Pyelonephritis is infection of the renal pelvis and parenchyma, usually from ascending infection. *E. coli* remains most common, but *Proteus*, *Klebsiella*, *Enterobacter*, *Pseudomonas*, *Enterococcus*, and resistant organisms are more likely when there is healthcare exposure, obstruction, stones, catheters, male sex, pregnancy, or recent antibiotics. The patient often looks systemically ill: fever, flank pain, CVA tenderness, nausea, vomiting, and tachycardia.

Pyelonephritis disposition algorithm

1. Confirm syndrome: fever or systemic symptoms plus flank pain/CVA tenderness and pyuria. Culture urine before antibiotics when possible.
2. Admit or observe if sepsis, hypotension, vomiting/inability to take PO, pregnancy, obstruction, transplant kidney, severe comorbidity, unreliable follow-up, or resistant-organism risk.
3. Image early if obstruction, stone, solitary kidney, renal transplant, emphysematous infection risk, persistent fever >48-72 hours despite therapy, or severe/recurrent disease.
4. Start empiric therapy based on local susceptibility and prior cultures. Narrow when results return.
5. If not improving, reconsider source control: obstructed infected stone, renal/perinephric abscess, emphysematous pyelonephritis, incorrect organism, or noninfectious mimic.

Emphysematous pyelonephritis is a gas-forming infection classically associated with diabetes or urinary obstruction and can be fatal without urgent antibiotics plus drainage or surgery. Xanthogranulomatous pyelonephritis, renal abscess, and perinephric abscess are chronic or complicated variants that often require imaging and procedural management.

VII. Prostatitis and male GU infection

Acute bacterial prostatitis is a systemic bacterial infection involving the prostate. It often presents with fever, chills, pelvic or perineal pain, dysuria, urinary frequency, obstructive symptoms, retention, and a tender boggy prostate. *E. coli* and other Enterobacterales predominate; risk increases after urinary instrumentation, catheterization, BPH, and urethral infection. Vigorous prostate massage is avoided because it can worsen bacteremia and pain.

Entity	Time course	Exam/lab pattern	Management
Acute bacterial prostatitis	Abrupt febrile illness	Tender boggy prostate; pyuria; urine culture often positive; blood cultures if toxic	Fluoroquinolone or TMP-SMX for mild outpatient cases when susceptible; IV broad therapy if septic, obstructed, vomiting, or high-risk; treat for weeks.
Chronic bacterial prostatitis	Recurrent UTIs with same organism, pelvic discomfort	Prostate may be normal or enlarged; cultures may localize organism	Prolonged antibiotic course with prostate penetration; recurrence common.
Chronic pelvic pain syndrome	Chronic pelvic pain without clear infection	Cultures often negative	Multimodal noninfectious management; avoid repeated antibiotics without evidence.

A man with UTI symptoms should not automatically be labeled as uncomplicated cystitis. Consider prostatitis, urethritis, obstruction, stones, instrumentation, and structural urinary disease. Recurrent male UTIs deserve evaluation for prostate, residual urine, stones, or anatomic abnormalities.

VIII. Sexually transmitted infections: syndromic approach

STI diagnosis is a clinical-pattern recognition task linked to public health. A patient may be asymptomatic, symptoms may overlap with UTI or vaginitis, and coinfection is common. The minimum safe framework is to identify the syndrome, test with NAAT or serology when appropriate, treat empirically when follow-up risk or complications are high, treat partners, and screen for other STIs including HIV and syphilis based on risk.

Syndrome	Likely causes	Diagnostic anchor	Treatment/management logic
Urethritis/cervicitis	Chlamydia trachomatis, Neisseria gonorrhoeae, Mycoplasma genitalium, Trichomonas in selected settings	NAAT from urine or genital site; test for both GC/CT	Doxycycline for chlamydia; ceftriaxone-based therapy for gonorrhea; partner treatment and abstinence until treatment complete.
Genital ulcer	HSV, syphilis, chancroid, lymphogranuloma venereum, trauma, Behcet, fixed drug eruption	HSV PCR/culture from lesion; syphilis serology; consider HIV testing	Treat HSV early; syphilis by stage; manage partners and pregnancy risks.
Pelvic inflammatory disease	Polymicrobial ascending infection; GC/CT common but not required	Clinical diagnosis: pelvic/lower abdominal pain plus cervical motion, uterine, or adnexal tenderness; test GC/CT/HIV/syphilis	Treat empirically to prevent infertility, ectopic pregnancy, chronic pelvic pain; include gonorrhea, chlamydia, anaerobes.
Genital warts	HPV, especially low-risk types 6 and 11	Usually clinical; biopsy atypical lesions	Patient-applied or clinician therapies; HPV vaccine prevention; screen cervix according to age/risk guidelines.

Chlamydia

Chlamydia trachomatis is a common bacterial STI and a frequent cause of urethritis and cervicitis. Many infections are asymptomatic. Men may have dysuria, urethral discharge, epididymitis, or proctitis. Women may have cervicitis, discharge, dysuria, intermenstrual bleeding, postcoital bleeding, pelvic pain, or PID. Untreated upper-tract disease can lead to infertility, ectopic pregnancy, and chronic pelvic pain. NAAT is the preferred diagnostic test in most settings.

Modern CDC guidance favors doxycycline 100 mg twice daily for 7 days for uncomplicated chlamydial infection in nonpregnant adults and adolescents, with azithromycin as an alternative in selected settings. Sexual partners should be evaluated and treated, and patients should abstain from sex until therapy is completed and partners are treated.

Gonorrhea

Gonorrhea can cause urethritis, cervicitis, PID, epididymitis, proctitis, pharyngitis, conjunctivitis, and disseminated gonococcal infection. Resistance drives treatment choices. Current U.S. treatment is ceftriaxone-based; add chlamydia treatment if chlamydia has not been excluded. Pharyngeal infection is harder to eradicate and needs test-of-cure in many protocols.

Syphilis

Syphilis is a staging diagnosis. Primary disease causes a chancre; secondary disease causes diffuse rash including palms/soles, condyloma lata, lymphadenopathy, fever, hepatitis, or neurologic/ocular findings; latent disease has serology without symptoms; tertiary disease includes gummas, aortitis, and neurologic syndromes. Diagnosis usually combines nontreponemal testing (RPR/VDRL, useful for titers) with treponemal confirmation. Penicillin remains the foundation, with regimen determined by stage and pregnancy status.

HIV-linked prevention reasoning

Every STI encounter is also a prevention encounter. HIV testing should be offered to patients being evaluated for STIs unless recently tested and low risk. Patients with bacterial STIs, recurrent STIs, injection drug use, or partners with HIV should be assessed for HIV pre-exposure prophylaxis (PrEP). A negative HIV test does not end prevention counseling if ongoing exposure risk persists.

IX. HIV and opportunistic infection bridge

This chapter focuses on GI/GU/STI infections, but these syndromes repeatedly intersect with HIV. Acute HIV may mimic mononucleosis or viral illness with fever, pharyngitis, rash, lymphadenopathy, diarrhea, aseptic meningitis, and mucocutaneous ulcers. Advanced HIV changes the infectious differential by CD4 count and by organ system. HIV testing is part of the diagnostic strategy for unexplained recurrent infections, severe or atypical infection, TB, certain STIs, chronic diarrhea, oral thrush, zoster in a young adult, and opportunistic presentations.

Immune marker	Infection pattern	Examples to keep active
Any CD4	Common infections may be more severe or recurrent	Bacterial pneumonia, HSV, VZV, TB, STIs

CD4 <200	Cell-mediated immunity failure	Pneumocystis jirovecii pneumonia; consider prophylaxis thresholds
CD4 <100	Toxoplasma and cryptococcal risk rises	Toxoplasma encephalitis in IgG-positive patients; cryptococcal meningitis
CD4 <50	Disseminated intracellular pathogens	MAC, CMV retinitis/colitis/esophagitis

The graph-RAG implication is important: HIV is not merely one node. It modifies the probability of TB, chronic diarrhea pathogens, severe HSV, CMV colitis, cryptococcal meningitis, recurrent bacterial pneumonia, non-Hodgkin lymphoma, Kaposi sarcoma, and medication-toxicity interactions. The same symptom, such as diarrhea or headache, activates a different diagnostic subgraph when CD4 count is low.

X. Diagnostic algorithms and high-yield comparison tables

Dysuria algorithm

1. Ask sex, pregnancy possibility, fever, flank pain, pelvic/perineal pain, discharge, lesions, new partners, and instrumentation/catheter history.
2. Classic cystitis without red flags: UA supports treatment; culture if recurrent, complicated, pregnancy, male sex, or treatment failure.
3. Dysuria plus urethral/cervical discharge or pyuria with negative culture: test GC/CT NAAT; consider M. genitalium, trichomonas, HSV, vaginitis.
4. Fever/flank/CVA tenderness: pyelonephritis pathway; culture and decide outpatient versus admission.
5. Male with fever/perineal pain/retention: prostatitis pathway; avoid prostate massage; culture; treat longer.

Diarrhea algorithm

1. Separate watery, inflammatory/bloody, fatty, and chronic diarrhea. Ask travel, antibiotics, hospitalization, food exposure, immunosuppression, sexual exposure, and outbreak contacts.
2. Watery diarrhea after antibiotics/hospitalization: test for CDI only if clinically significant unformed stools.
3. Bloody diarrhea, high fever, severe abdominal pain, sepsis, immunocompromise, or prolonged symptoms: stool testing and targeted therapy; avoid antimotility drugs when invasive colitis suspected.
4. Chronic diarrhea with HIV risk or low CD4: broaden to Giardia, Cryptosporidium, Microsporidia, CMV colitis, MAC, and medication effects.
5. Persistent focal abdominal pain, fever, or post-procedural illness: image for abscess or source-control problem.

Presentation	Most likely region	Must-not-miss
Watery diarrhea after clindamycin/cephalosporin/fluoroquinolone	Colon	C. difficile colitis, toxic megacolon
RUQ pain + fever + jaundice	Biliary tree	Ascending cholangitis needing decompression
Dysuria/frequency without fever	Bladder/urethra	STI if discharge or sterile pyuria
Flank pain + fever + pyuria	Kidney	Obstruction, renal abscess, sepsis
Perineal pain + fever + tender prostate	Prostate	Bacteremia, urinary retention, abscess
Descending paralysis after preserved food	Neuromuscular junction	Botulism with respiratory failure
Jaundice + INR elevation + confusion	Liver	Acute liver failure

XI. Initial management patterns

Initial infectious-disease management should be framed as three simultaneous tasks: stabilize the patient, identify the source, and start empiric therapy when delay is dangerous. Cultures are useful when they can be obtained promptly and safely, but unstable patients should not wait for an exhaustive diagnostic workup before antibiotics and source control.

Clinical situation	Immediate steps	Common pitfall
Sepsis with suspected intra-abdominal source	Blood cultures if no delay, lactate, fluids/vasopressors as needed, broad antibiotics, urgent CT/surgery/interventional radiology for source control	Giving antibiotics but missing perforation, abscess, ischemia, or obstruction

Fulminant CDI	Oral/NG vancomycin plus IV metronidazole, rectal vancomycin if ileus, stop inciting antibiotics, early surgical consult	Waiting for stool output/testing when ileus prevents diarrhea
Pyelonephritis with obstruction	IV antibiotics, urgent decompression of infected obstructed system	Treating with antibiotics alone while infected urine remains blocked
Acute prostatitis with retention	Antibiotics, bladder drainage strategy, avoid prostate massage	Vigorous exam/massage causing bacteremia
Botulism	Airway monitoring, public health notification, antitoxin, ICU if bulbar or respiratory signs	Waiting for confirmatory toxin assay before treatment
Suspected acute liver failure	Check INR/glucose/mental status, stop hepatotoxins, acetaminophen level, N-acetylcysteine when appropriate, urgent hepatology/transplant consultation	Reassuring oneself based only on transaminases

XII. Practice questions

Question 1

A 72-year-old hospitalized patient develops 8 watery stools/day after ceftriaxone. WBC is 18,000 and creatinine rose from 0.9 to 1.6. Abdomen is tender but not distended; no shock. What is the diagnosis category and first-line treatment logic?

Answer: Severe nonfulminant *C. difficile* infection. Stop inciting antibiotics when possible, use fidaxomicin if feasible or oral vancomycin as acceptable therapy, monitor for fulminant features. Do not use antimotility agents as a substitute for treatment.

Question 2

A 24-year-old woman has dysuria and frequency without fever or flank pain. UA shows leukocyte esterase and nitrites. She is not pregnant. What is the likely syndrome and what changes the workup?

Answer: Uncomplicated cystitis. Culture/workup expands if symptoms are recurrent, atypical, persistent after therapy, pregnancy, pyelonephritis signs, immunocompromise, or concern for STI.

Question 3

A 45-year-old man has fever, chills, dysuria, urinary retention, and severe perineal pain. DRE reveals a tender boggy prostate. What should be avoided and why?

Answer: Avoid vigorous prostate massage because acute bacterial prostatitis can seed bacteremia and worsen pain. Treat with antibiotics and manage retention/sepsis risk.

Question 4

A patient has HBsAg positive, anti-HBc IgM positive, anti-HBs negative. What does this mean?

Answer: Acute hepatitis B infection. HBsAg indicates infection, anti-HBc IgM indicates recent/acute infection, and anti-HBs has not yet developed.

Question 5

A patient has a positive HCV antibody but undetectable HCV RNA. What does this mean clinically?

Answer: It suggests prior exposure with no current active viremia, or rarely false-positive antibody. Active HCV infection requires detectable RNA.

Question 6

A patient develops diplopia, dry mouth, dysarthria, and descending weakness after eating home-canned food. What management action should not wait for lab confirmation?

Answer: Botulism antitoxin and airway monitoring/public health involvement should not wait for confirmatory toxin testing.

Question 7

A woman with pelvic pain has cervical motion tenderness and mucopurulent discharge. Pregnancy test is negative. Why treat empirically?

Answer: PID is a clinical diagnosis and delayed treatment increases risk of infertility, ectopic pregnancy, and chronic pelvic pain. Treat for gonorrhea, chlamydia, and anaerobic/polymicrobial pathogens and evaluate partners.

XIII. Graph-RAG extraction map

The following concept map is designed for graph-RAG ingestion. Nodes should store definitions, clinical features, diagnostic criteria, management anchors, complications, and source-context links. Edges should capture causal mechanisms, diagnostic discriminators, treatment links, and contraindication relationships.

Node	Class	High-value facts	Edges to activate
Viral hepatitis	GI infection / hepatology	HAV/HEV fecal-oral; HBV blood/sex/perinatal; HCV blood; HBV/HCV/HDV chronic risk; acute liver failure defined by encephalopathy and coagulopathy	causes -> hepatocellular injury; diagnosed by -> serology/RNA; complicated by -> cirrhosis/HCC; prevented by -> vaccines for HAV/HEV
C. difficile infection	GI infection / antibiotic-associated colitis	Watery diarrhea after microbiome disruption; severe by WBC/Cr; fulminant with shock/ileus/megacolon	caused by -> antibiotics/hospitalization; treated by -> fidaxomicin/vancomycin; complicated by -> toxic megacolon; prevented by -> stewardship/contact precautions
Intra-abdominal abscess	Source-control infection	Polymicrobial; CT/US localization; antibiotics plus drainage	requires -> source control; covers -> gram-negatives/anaerobes/enterococci; mimics -> persistent fever after abdominal process
Uncomplicated cystitis	GU infection	Dysuria/frequency/urgency; no fever/flank/systemic signs; E. coli most common	distinguished from -> pyelonephritis/STI/vaginitis; treated by -> nitrofurantoin/TMP-SMX/fosfomycin; diagnosed by -> UA
Pyelonephritis	Upper GU infection	Fever, flank pain, CVA tenderness, pyuria; culture; image if obstruction or no response	complicated by -> sepsis/abscess; treated by -> empiric antibiotics; requires -> admission if severe
Prostatitis	Male GU infection	Acute form causes fever, pelvic pain, tender boggy prostate; chronic form causes recurrent UTIs	avoid -> prostate massage; treated by -> prolonged prostate-penetrating antibiotics; complicated by -> retention/abscess
Chlamydia	STI	Often asymptomatic; urethritis/cervicitis/PID/epididymitis; NAAT; doxycycline preferred for many nonpregnant adults	coinfects with -> gonorrhea; causes -> PID/infertility/reactive arthritis; requires -> partner treatment
HIV modifier	Host-risk axis	Low CD4 broadens differential; STI visit is HIV testing/prevention opportunity	modifies -> TB/diarrhea/CNS/STI differential; prevents via -> PrEP/ART; complicated by -> opportunistic infections
Botulism	Toxin-mediated neuroinfection	Descending paralysis, cranial nerve signs, preserved sensation; airway risk; antitoxin	caused by -> preformed toxin/wound/infant colonization; mimics -> GBS/myasthenia; treated by -> antitoxin/supportive ventilation

XIV. Common pitfalls

- Treating asymptomatic bacteriuria in most nonpregnant adults, especially older or catheterized patients.
- Testing for C. difficile in formed stool or repeating tests as a cure check.
- Forgetting that nitrite-negative UA does not rule out UTI, especially with Enterococcus, Staph saprophyticus, frequent urination, or low bacterial burden.
- Missing STI-related urethritis in a patient labeled as recurrent UTI.
- Using antibiotics alone for an abscess, obstructed infected kidney, or cholangitis without addressing drainage/source control.
- Ignoring acute liver failure markers because the patient is still early in the course of viral hepatitis.
- Failing to treat sexual partners or to counsel abstinence until STI treatment is complete.
- Not considering HIV testing and PrEP counseling during STI, TB, recurrent infection, or unexplained chronic diarrhea encounters.

Guideline and reference anchors used for clinical cross-checking

External clinical anchors consulted while writing this original chapter included IDSA/SHEA guidance for *C. difficile* infection, CDC sexually transmitted infection treatment guidance, AASLD/IDSA hepatitis C guidance, AASLD hepatitis B guidance, CDC viral hepatitis resources, CDC/NIH/HIVMA opportunistic infection resources, and IDSA-style syndrome management principles for complicated intra-abdominal and urinary infections. The uploaded source section was used for topic coverage and sequencing, not copied wording.